

Functional Annotation Report: *accD* (Q88LD9) — Acetyl-CoA carboxylase carboxyltransferase β -subunit

Organism: *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440), abbrev. **PSEPK**
Gene: *accD* — OrderedLocusName **PP_1996** **Protein:** Acetyl-coenzyme A carboxylase carboxyltransferase subunit β (ACCase subunit β) **EC number:** 2.1.3.15 **Protein family:** AccD/PCCB family **UniProt:** Q88LD9

1. Summary (Answer to the Research Question)

accD encodes the **β -subunit of the carboxyltransferase (CT) component of acetyl-CoA carboxylase (ACC)**, a large biotin-dependent multi-subunit enzyme that catalyzes the **first committed and rate-limiting step of fatty-acid (and membrane phospholipid) biosynthesis**: the ATP-dependent carboxylation of acetyl-CoA to **malonyl-CoA** (overall EC 6.4.1.2; the CT partial reaction is EC 2.1.3.15). In bacteria the holoenzyme is built from four separately encoded proteins — **AccC** (biotin carboxylase, BC), **AccB** (biotin carboxyl carrier protein, BCCP), and the two carboxyltransferase subunits **AccA** (α) and **AccD** (β). AccA and AccD assemble into an **$\alpha_2\beta_2$ heterotetramer** that catalyzes the **second half-reaction**: transfer of the carboxyl group from carboxybiotin (carried on BCCP) onto acetyl-CoA to form malonyl-CoA [PMID 23594205; 19965770]. AccD's substrate is acetyl-CoA (with carboxybiotin-BCCP as the CO₂ donor), and its product is malonyl-CoA. The enzyme is a **soluble cytoplasmic** protein. A defining, **bacteria-specific zinc-binding domain** resides on the β /AccD subunit, making the bacterial CT a validated selective antibacterial drug target [PMID 16460018].

Identity verification. The UniProt description (*accD*; carboxyltransferase β -subunit; EC 2.1.3.15; AccD/PCCB family; domains COA_CT_N, AcCoA_carboxyl_beta, ClpP/crotonase-like fold, Znf-ACC) matches the well-characterized bacterial ACC carboxyltransferase β -subunit exactly. All literature cited below concerns this enzyme family in Gram-negative γ -proteobacteria (*E. coli*, *Haemophilus*, *Pseudomonas*), of which *P. putida* KT2440 is a member. The gene symbol is **not** ambiguous. Because no primary study is devoted

specifically to *P. putida* PP_1996, its function is assigned by **strong homology/HAMAP rule MF_01395**, with the mechanism established biochemically and structurally in closely related bacteria.

2. Primary Function: The Catalyzed Reaction and Substrate Specificity

2.1 Overall reaction

Acetyl-CoA carboxylase catalyzes:



This is **the first committed step in fatty-acid synthesis** and is the pathway's principal regulatory/rate-limiting valve [PMID 29100983; 39572150].

2.2 Two half-reactions — AccD acts in the second

The biotin-dependent mechanism proceeds in two spatially separated half-reactions [PMID 23594205; 29100983; 26020841]:

1. **Half-reaction 1 (biotin carboxylase, AccC + AccB):** ATP-dependent carboxylation of the vitamin **biotin**, which is covalently attached to a lysine of **BCCP (AccB)**, using **bicarbonate** as the CO₂ source → **carboxybiotin-BCCP**.
2. **Half-reaction 2 (carboxyltransferase, AccA + AccD + AccB):** **AccD together with AccA** catalyzes transfer of the carboxyl group from **carboxybiotin-BCCP** onto **acetyl-CoA**, yielding **malonyl-CoA** and regenerating biotin-BCCP.

AccD is one of the two catalytic subunits of the carboxyltransferase (CT) active site. The CT is an **α₂β₂ heterotetramer** (α = AccA, β = AccD); its two active sites are each formed at an **α/β subunit interface**, and the two sites are functionally coupled ("active-site communication") [PMID 19965770; 18156466; 23594205].

2.3 Substrate specificity

- **Carboxyl acceptor / defining substrate: acetyl-CoA** (a 2-carbon acyl-CoA). The bacterial ACC is specific for acetyl-CoA (distinguishing it from the homologous propionyl-CoA carboxylases of the AccD/PCCB family that act on propionyl-CoA).
- **Carboxyl donor: carboxybiotin** tethered to BCCP (AccB).
- **Product: malonyl-CoA** (+ regenerated biotin-BCCP).
- The reaction is readily reversible in vitro (malonyl-CoA + biotin $\rightleftharpoons$ acetyl-CoA + carboxybiotin), which is the basis of standard CT activity assays [PMID 16460018; 39572150].

3. Structure and Structural Basis of Function

3.1 Fold and domain architecture

The CT subunits (AccA and AccD) belong to the **ClpP/crotonase-like superfamily fold** (InterPro IPR029045), the conserved "CoA carboxyltransferase" (CT) fold shared with the β -subunits of propionyl-/methylcrotonyl-CoA carboxylases (AccD/PCCB family) [UniProt Q88LD9 domains]. The Q88LD9 domain complement — **COA_CT_N (IPR011762)**, **Acetyl_CoA_COase_Trifase_b_su (IPR000438)**, **AcCoA_carboxyl_beta (IPR034733)** — is the canonical β -CT signature.

3.2 The bacteria-specific zinc domain (a key discovery)

High-resolution crystal structures of the CT component from *E. coli* (3.0 Å) and *Staphylococcus aureus* (2.0 Å) revealed a **small, independent zinc-binding domain that has no counterpart in the eukaryotic ACC** [PMID 16460018]. This zinc finger: - resides on the **β -subunit (AccD)** and is coordinated by **four cysteines** (a Cys₄ Zn site; corresponds to the Q88LD9 annotation **Znf-ACC, IPR041010**); - is **required for catalytic activity** — mutation of the four cysteines to alanine abolishes nucleic-acid binding **and** diminishes catalysis, indicating the Zn domain is important for CT structural integrity/function [PMID 19965770].

The structural distinctiveness of this domain is the central reason bacterial CT (and the AccD subunit specifically) is repeatedly highlighted as a **selective antibacterial target** that spares the human enzyme [PMID 16460018; 23499019; 39572150].

3.3 Assembly into the holoenzyme

Individual bacterial ACC subunits (BC homodimer, BCCP, and the $\alpha_2\beta_2$ CT) associate into a **multimeric holo-complex**; pull-down and kinetic studies in *E. coli* show that the two catalytic sites (BC and CT) **interact** and that complex formation occurs (and is not abolished by acetyl-CoA, ATP analogs, or mRNA) [PMID 23594205; 23499019]. Thus AccD functions **in the context of an assembled ACC complex**, not as an isolated protein.

4. Subcellular Localization

In bacteria, acetyl-CoA carboxylase — including the AccA/AccD carboxyltransferase — is a **soluble, cytoplasmic enzyme complex** [PMID 23594205]. Catalysis (malonyl-CoA formation) occurs in the **cytosol**, where malonyl-CoA is then handed to the soluble type II fatty-acid-synthase (FAS II) machinery. (This contrasts with plants, where the heteromeric ACCase is plastid-localized and its α -CT associates with the plastid inner-envelope membrane [PMID 39489480]; and with the cytosolic multidomain ACC of mammals.) For *P. putida*, a Gram-negative γ -proteobacterium, the *E. coli* cytoplasmic paradigm applies.

5. Pathway Context and Biological Process

5.1 Position in metabolism

AccD/ACC sits at the **entry point of type II fatty-acid biosynthesis (FAS II)**:

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acetyl-CoA → (ACC: AccABCD) → malonyl-CoA → (FabD: malonyl-CoA:ACP transacylase) →
malonyl-ACP → iterative condensation/reduction (FabB/F, FabG, FabA/Z, FabI) → acyl-ACP
→ membrane phospholipids
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Malonyl-CoA produced by the AccD-containing CT is the **universal two-carbon extender unit** for every elongation cycle of fatty-acid synthesis and is therefore essential for **membrane phospholipid biogenesis** [PMID 19362649; 8246839]. In many bacteria malonyl-CoA is additionally the extender for **polyketide** biosynthesis, linking ACC to secondary metabolism [PMID 19362649; 41203165].

5.2 Regulation of the step

Because this is the committed step, it is tightly controlled: - **Product/end-product feedback:** ACC is allosterically inhibited by long-chain **acyl-ACP** (e.g., palmitoyl-ACP), which acts in a time-dependent (hysteretic) manner — coupling ACC output to the cell's demand for acyl chains [PMID 29100983]. - **Metabolic sensing / proposed moonlighting:** The CT (AccA/AccD) was proposed to act as a "dimmer switch," binding *accA/accD* mRNA through the β -subunit zinc finger to repress its own translation, an effect relieved by the substrate acetyl-CoA [PMID 19965770]. **However, subsequent in vivo work found no detectable effect of excess CT tetramer on translation of the CT mRNAs, arguing against physiologically significant translational repression** [PMID 25157077]. Nonspecific nucleic-acid (DNA/RNA/heparin) binding by CT also inhibits catalysis in vitro and was used to probe active-site communication [PMID 18156466]. The **primary, well-supported role of AccD is enzymatic** (carboxyl transfer); the regulatory RNA-binding role remains unresolved. - **Transcriptional control:** dedicated activators tuning *acc* gene expression in response to acetyl-CoA have been described in other γ -proteobacteria (e.g., AccR in *Shewanella*) [PMID 39475184].

5.3 Gene organization in *Pseudomonas*

In *P. aeruginosa* (a close relative), **accB and accC form a two-gene operon**, while **accA and accD are dispersed elsewhere on the genome**, not adjacent to accBC — mirroring the four separately located *acc* loci of *E. coli* [PMID 7693652]. The *P. putida* KT2440 locus tag **PP_1996** for *accD* is consistent with this dispersed arrangement. This physical separation of the four subunit genes creates a **subunit-stoichiometry coordination problem** that motivated the (now-disputed) autoregulation hypothesis above.

6. Essentiality and Significance as a Drug Target

Fatty-acid synthesis is **essential** in most bacteria, and ACC catalyzes its committed step; consequently AccD/ACC is essential for viability and growth. The bacterial CT's **unique zinc domain and the two-component (BC + CT) architecture** — absent from the single-chain eukaryotic ACC — make it an actively pursued **antibacterial (and herbicidal/insecticidal in eukaryotic ACC contexts) drug target** [PMID 16460018; 23499019; 36459397; 39572150]. Comparative structural/in-silico analyses continue to nominate bacterial ACCase as a target across Gram-positive and Gram-negative pathogens [PMID 42379113].

7. Evidence Summary and Confidence

Claim	Evidence type	Confidence	Key refs
accD = β -subunit of ACC carboxyltransferase ($\alpha_2\beta_2$)	Genetics + biochemistry (<i>E. coli</i>); homology for PSEPK	High	19965770, 23594205
Catalyzes 2nd half-reaction: carboxybiotin + acetyl-CoA $\rightarrow$ malonyl-CoA	Enzyme kinetics, mechanism	High	23594205, 29100983
Substrate = acetyl-CoA; product = malonyl-CoA	Biochemistry	High	29100983, 16460018
First committed / rate-limiting step of FAS	Biochemistry, review	High	29100983, 39572150
Bacteria-specific Cys ₄ Zn domain on β -subunit, needed for catalysis	X-ray crystallography + mutagenesis	High	16460018, 19965770
Cytoplasmic, soluble multi-subunit complex	Biochemistry / pull-downs	High	23594205
Feedback inhibition by acyl-ACP	Kinetics	High	29100983
accA/accD unlinked from accBC in <i>Pseudomonas</i>	Cloning/hybridization (<i>P. aeruginosa</i>)	Medium-High (relative)	7693652
CT mRNA-binding autoregulation ("dimmer switch")	In vitro (proposed) vs in vivo (refuted)	Contested	19965770 (for) / 25157077 (against)
PSEPK-specific functional assignment	Homology / HAMAP MF_01395	Inferred (by similarity)	UniProt Q88LD9

8. Supported and Refuted Hypotheses

Supported - AccD is the β -subunit of the carboxyltransferase that produces malonyl-CoA (2nd half-reaction of ACC). ✓ - The enzyme is cytoplasmic and functions within an assembled multi-subunit ACC complex. ✓ - AccD bears a bacteria-specific Cys₄ zinc domain required for catalysis, marking it as a selective drug target. ✓ - The reaction is the committed, feedback-regulated (acyl-ACP) entry step of fatty-acid/phospholipid biosynthesis. ✓

Refuted / Contested - CT (AccA/AccD) acts as a physiologically significant translational repressor of its own mRNAs — **not supported in vivo** [PMID 25157077], despite an in-vitro "dimmer switch" proposal [PMID 19965770]. AccD's principal role is enzymatic; any nucleic-acid-binding/moonlighting function is unresolved and likely minor.

9. Limitations and Future Directions

- **No primary study on PP_1996 itself.** The functional assignment for *P. putida* KT2440 rests on strong homology (HAMAP MF_01395) and mechanism established in *E. coli*, *S. aureus*, and *Haemophilus*. Direct biochemical/structural confirmation of the KT2440 protein is lacking.
 - **Structural specifics of the *Pseudomonas* CT** (e.g., a species structure, exact Zn-domain geometry) have not been reported; a *P. putida* CT crystal/cryo-EM structure would confirm domain conservation.
 - **Regulatory role of the AccD zinc domain in RNA binding** remains contested and merits definitive in vivo testing in Pseudomonads.
 - **Subunit stoichiometry coordination** from dispersed *acc* loci in *Pseudomonas* is mechanistically unresolved.
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Key References (PMID)

- 16460018 — Bilder et al. (2006) *Biochemistry*. Structure of bacterial CT reveals the unique zinc-binding motif.
- 19965770 — Meades et al. (2010) *Nucleic Acids Res.* CT enzymatic activity + proposed translational repression via β -subunit zinc finger.

- 25157077 — Smith & Cronan (2014) *J. Bacteriol.* In vivo evidence against CT translational repression.
- 23594205 — Broussard et al. (2013) *Biochemistry*. Complex formation and interacting active sites of *E. coli* ACC.
- 23499019 — Broussard et al. (2013) *Structure*. BC–BCCP complex structure.
- 29100983 — Evans et al. (2017) *Biochim. Biophys. Acta*. Hysteretic acyl-ACP feedback inhibition of ACC.
- 18156466 — Benson et al. (2008) *Biochemistry*. DNA inhibits CT; active-site communication.
- 26020841 — Broussard et al. (2015) *Biochemistry*. Biotin carboxylase mechanism/structures.
- 7693652 — Best & Knauf (1993) *J. Bacteriol.* Genetic organization of *P. aeruginosa* ACC (accBC operon; accA/accD dispersed).
- 19362649 — Cronan & Thomas (2009) *Methods Enzymol.* Bacterial fatty-acid synthesis (review).
- 8246839 — Magnuson et al. (1993) *Microbiol. Rev.* Regulation of *E. coli* fatty-acid biosynthesis (review).
- 39572150 — Čavuzić et al. (2024). Methodology of ACC inhibition (review).